

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0504

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	Question	Bloom's Level
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

Code of Conduct:

I B Aishwarya/192421432 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B Aishwarya

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

Answers:

1.Schema

A schema is the overall logical design or structure of a database. It defines what data is stored, the tables/entities, their attributes, data types, and the relationships and constraints among them. A schema is defined at the time the database is designed and does not change frequently — it changes only when the database structure itself is altered.

Example: A schema for a college database may define

- Student (Roll No, Name, Department, Year)
- Course (Course ID, Course Name, Credits)

This describes the structure — it does not show any actual student names or marks.

2. Instance

An instance (also called a database state or snapshot) is the actual data stored in the database at a specific moment in time. Unlike the schema, the instance changes very frequently as records are inserted, updated, or deleted.

Example: At 10:00 AM the student table may contain the row (101, "Arun", "CSE", 2), and after a new admission at 11:00 AM it may also contain (102, "Divya", "IT", 1). Both rows together form the current instance of the student table.

3. Logical, Physical, and View Schema

The architecture separates a database into three levels, each with its own schema, so that changes at one level do not affect the others.

Aspect	Describes	Example
Physical Schema (Internal level)	How data is actually stored on disk — file organization, indexing, storage allocation, and access paths.	Student data stored as a B-Tree indexed file on Roll No.
Logical Schema (Conceptual level)	What data is stored and the relationships/constraints among	Student table linked to Course table through Enrolment, with Roll No as primary key.

	them, independent of physical storage details.	
View Schema (External level)	The part of the database that is relevant to a particular user or application; different users can have different views of the same data.	An Accounts clerk sees only fee-related fields, while a faculty member sees only marks and attendance fields.

Q2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Answers:

1. Entities and Attributes

Entity	Key Attribute	Other Attributes
CUSTOMER	Cust ID	Name, Address, Phone
ACCOUNT	Acc No	Account Type, Balance
BRANCH	Branch ID	Branch Name, City
LOAN	Loan ID	Loan Type, Amount
EMPLOYEE	Emp ID	Name, Designation

2. Relationships and Cardinalities

- CUSTOMER HAS ACCOUNT — one customer can hold many accounts, but each account belongs to one customer (1: M).
- ACCOUNT MAINTAINED BY BRANCH — many accounts are maintained by one branch (M: 1).
- CUSTOMER AVAILS LOAN — one customer can avail many loans (1: M).
- BRANCH OFFERS LOAN — one branch offers many loans (1: M).
- EMPLOYEE WORKS AT BRANCH — many employees work at one branch (M: 1).

3. ER Diagram

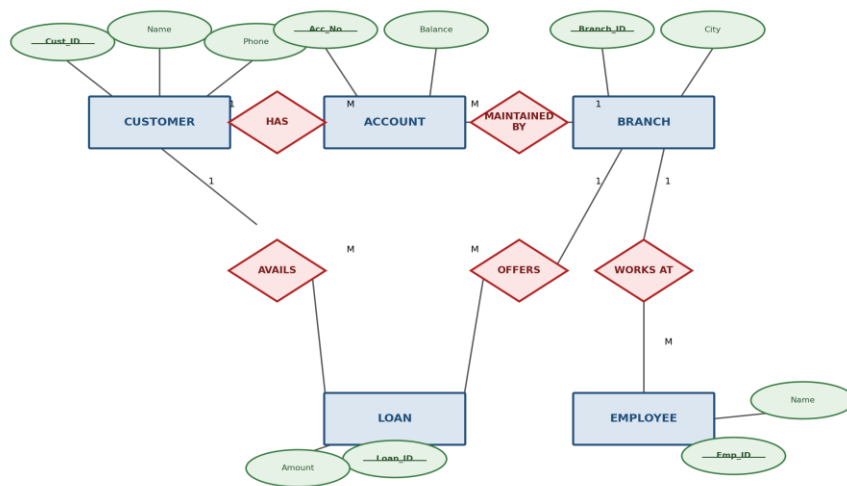


Figure 1: ER Diagram for Bank Management System

Q3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.

Answers:

1. Extended ER (EER) Model

The basic ER model is good for representing simple, flat entities and relationships, but many real-world applications need to model entities that have sub-types, shared characteristics, or relationships that themselves participate in further relationships. The Extended ER (EER) model adds three additional constructs to the basic ER model — generalization, specialization, and aggregation — along with concepts such as superclass/subclass and inheritance, to model such complex data requirements more accurately.

2. Generalization

Generalization is a bottom-up process in which two or more lower-level entities that share common attributes are combined to form a single, higher-level generalized entity (superclass). It emphasises the similarities between entity types and hides their differences.

Example: The entities CAR and TRUCK, which share attributes such as Vehicle No and Registration Date, can be generalized into a single higher-level entity VEHICLE.

3. Specialization

Specialization is the reverse, top-down process in which a higher-level entity is divided into two or more lower-level entities (subclasses) based on some distinguishing characteristic. Each subclass inherits all the attributes and relationships of the superclass and additionally has its own specific attributes.

Example: The entity EMPLOYEE can be specialized into ENGINEER, MANAGER, and CLERK, where each subclass has additional attributes of its own, such as Project Handled for Engineer or Team Size for Manager, while still inheriting Emp ID and Name from Employee.

Specialization/generalization is further classified using two constraints:

- Disjoint (d) vs Overlapping (o): whether an entity instance can belong to only one subclass (disjoint) or more than one subclass at the same time (overlapping).
- Total vs Partial: whether every instance of the superclass must belong to some subclass (total) or only some instances need to (partial).

4. Aggregation

Aggregation is an EER concept that allows a relationship between entities to be treated as a higher-level entity in its own right, so that this relationship can itself participate in another relationship with a different entity. It is used when a relationship needs to have its own relationship with another entity, which the basic ER model cannot express directly.

Example: The relationship WORKS_ON between EMPLOYEE and PROJECT can be aggregated into a single higher-level construct so that it can further participate in a relationship, such as USES, with the entity EQUIPMENT — representing which equipment is used for a given employee-project assignment.

Q4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

Answers:

1. Entities Identified

- PERSON (superclass) — Person ID, Name, Address
- PATIENT (subclass of Person) — Patient ID, Diagnosis
- DOCTOR (subclass of Person) — Doctor ID, Specialization
- NURSE/STAFF (subclass of Person) — Staff ID, Shift

- DEPARTMENT — Dept ID, Dept Name
- APPOINTMENT (weak entity, existence-dependent on PATIENT) — Appt No (partial key), Date/Time

2. Inheritance

PERSON is modelled as a superclass that is specialized (disjoint, shown by the circled "d") into three subclasses — PATIENT, DOCTOR, and NURSE/STAFF. Each subclass inherits Person ID, Name, and Address from PERSON and adds its own specific attributes, avoiding repetition of common attributes across the three subclasses.

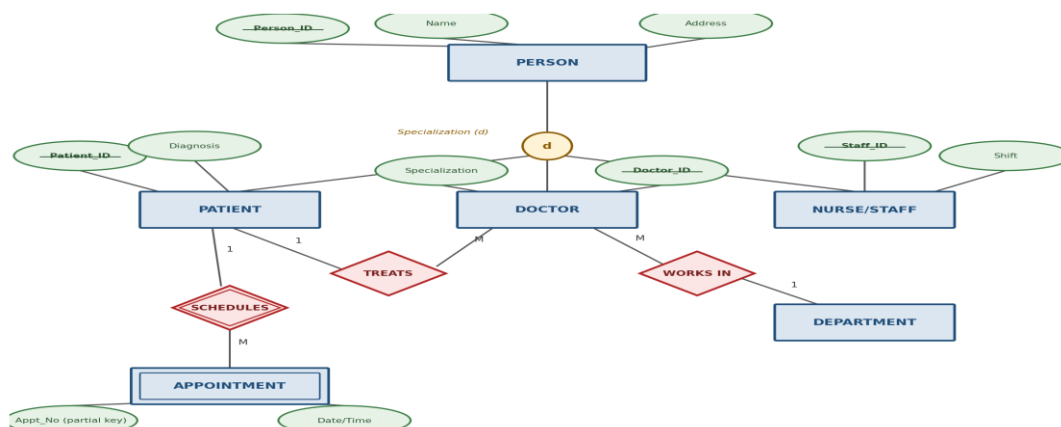
3. Weak Entity

APPOINTMENT is a weak entity because it has no independent primary key of its own — an appointment cannot exist without the PATIENT it belongs to. It is identified only in combination with the PATIENT entity through the identifying relationship SCHEDULES, using its partial key Appt_No. In the diagram, the weak entity and its identifying relationship are shown with double-lined borders.

4. Other Relationships

- DOCTOR TREATS PATIENT (M : 1) — one doctor treats many patients.
- DOCTOR WORKS IN DEPARTMENT (M : 1) — many doctors work in one department.
- PATIENT SCHEDULES APPOINTMENT (1 : M, identifying relationship for the weak entity).

5. EER Diagram



Double-lined diamond/box = identifying relationship / weak entity | Circle with 'd' = disjoint specialization

Figure 2: EER Diagram for Hospital Management System

Q5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Answers:

A Database Administrator (DBA) is the person, or team of persons, responsible for the overall control, management, and upkeep of a database management system and its data. The DBA acts as the custodian of the data, ensuring it remains accurate, secure, and available to authorized users at all times.

Key Responsibilities

- **Schema Definition:** Creates the original database schema by defining tables, attributes, relationships, and constraints (using DDL statements).
- **Storage Structure and Access Method Definition:** Decides how data is physically stored on disk and which access/indexing methods are used for efficient retrieval.
- **Schema and Physical Organization Modification:** Alters the logical schema or physical storage structure as organizational requirements evolve, with minimal disruption to existing applications.
- **Granting Authorization for Data Access:** Controls which users or roles can access, insert, update, or delete specific parts of the data, enforcing access control policies.
- **Routine Maintenance:** Performs regular backups, monitors performance, applies patches/upgrades, and tunes queries and indexes.
- **Ensuring Data Security:** Protects the database against unauthorized access, SQL injection, and other threats through authentication and encryption mechanisms.
- **Ensuring Data Integrity:** Enforces integrity constraints (primary key, foreign key, check constraints) so that the data remains accurate and consistent.
- **Backup and Recovery:** Establishes backup schedules and recovery procedures so the database can be restored after failures, crashes, or disasters.
- **Coordinating with Users:** Liaises with end-users and application developers to understand new requirements, resolve conflicts over data usage, and provide training.